

MALLIKARJUNA K

Dallas, Tx (Open to Relocation) | +1(972) 924-0695 | mallikarjunakovi45@gmail.com

SUMMARY:

AI Software Engineer with 4+ years of experience (inclusive of advanced research) in designing and deploying enterprise-grade **Semantic Search** and **GenAI RAG platforms**. Expert in Python-based OOP, **FastAPI microservices**, and **MLOps** lifecycle management. Proven track record of automating CI/CD toolstacks and mentoring teams in Agile environments. Specialized in building secure, scalable AI systems featuring **JWT integration**, **API Gateways**, and automated testing frameworks (**pytest**, **SonarQube**).

PROFESSIONAL EXPERIENCE

Citibank | GenAI Developer | Irving, Tx Jan 2025-Current

- Accomplished sub-2-second retrieval latency (X) over 10k+ compliance documents (Y) by designing a production-ready RAG framework using FastAPI and Semantic Search with Hybrid Reranking (Z).
- Mentored a team of 4 junior engineers on CI/CD best practices, standardizing a shared multi-repo codebase in VS Code and automating toolstacks via Jenkins and SonarQube to ensure 100% code quality compliance.
- Reduced LLM hallucinations by 30% as measured by the RAGAS framework by implementing advanced chunking strategies and context filtering within a secure API Gateway architecture.
- Architected a secure authentication layer using JWT Integration, ensuring the LLM only generates responses based on user-specific RBAC clearance in a highly regulated banking environment.
- Executed spikes and Proofs of Concept (PoCs) for MCP (Model Control Plane) modules, resulting in the successful integration of open-source models for internal summarization use cases.
- Automated manual release activities by developing Ansible playbooks for microservice deployment, maintaining 99.9% uptime for internal compliance tools.

Infosys | Hyderabad, India Jan 2022-Dec 2023
Software Developer

- Accomplished a 22% improvement in inventory replenishment accuracy by developing an OOP-based demand forecasting engine using Python, LSTM, and Prophet, delivered through a robust FastAPI service.
- Optimized data pipelines for millions of records using Pandas and NumPy, reducing preprocessing time by 40% through automated feature engineering and Data & Analytics validation.
- Engineered automated test suites (Integration and Regression) using pytest, achieving 90% code coverage and mitigating production risks for a high-volume e-commerce client.
- Participated in story refinement and estimation in an Agile environment, collaborating with Data Scientists and Product Teams to translate complex business requirements into scalable architectural components.
- Managed multiple priorities across cross-functional Sales and Finance teams, delivering a business intelligence dashboard that visualized model risk and forecasting KPIs.

Web Developer (Intern) Mar 2021-Jan 2022

- Designed and developed a Department of Information Technology website using HTML, CSS, and JavaScript to present courses, faculty profiles, research activities, and announcements.
- Implemented interactive and dynamic features including navigation menus, event calendars, and contact forms to improve usability for students and staff.

- Skills
- AI/GenAI:** RAG (Chunking, Embedding, Reranking, Summarization), Fine-Tuning, Semantic Search, Open-source LLMs, MCP Modules, MLOps, LangChain, LlamaIndex.
 - Software Development:** Python (Expert), Java, Scala, FastAPI, RESTful APIs, JWT, API Gateway (Kong/Apigee), OOP, Design Patterns, VS Code (Multi-repo workflows).
 - Data & Analytics:** MongoDB, Redis, SQL (PostgreSQL/MS SQL), Apache Kafka, Data Pipelines, Feature Engineering, Vector Databases (Pinecone, Milvus).
 - DevOps & CI/CD:** Jenkins, Ansible, Docker, Kubernetes, Artifactory, SonarQube, Checkmarx, GIT/Bitbucket, Terraform.
 - Testing & Quality:** Pytest, Unit/Integration/Regression/Performance Testing, Secure SDLC.

Education

University of Cincinnati – Masters in Information Technology | GPA: 4.0 Jan 2024- Apr 2025